

DR-2 incremental lemmas: the diametral-disjointness promotion, the extraction decomposition, and the exact sticking point

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Verdict #14 kept the unrestricted-class closure open via DR-2 (research-grade, publication-strength alternative) and the DR-2 seed lemma (pigeonhole) was registered the same day. This note formalizes the seed into an extraction architecture, proving the two exact pieces and NAMING the precise obstruction where the elementary route stops. Scope: the unrestricted single-shell class (no H-ADM-COH); pure combinatorics of the carrier partition; off the critical path by verdict #14.

2. The lemmas

Lemma 1 (L1: diametral-disjointness promotion – exact). *On a circle D , every point has exactly one antipode (within D), so distinct antipodal (diametral) pairs of D are DISJOINT two-point sets. Hence if a carrier D carries p_D pattern antipodal pairs, then $|D \cap P| = 2p_D$ exactly: a carrier rich in pairs IS a pattern circle rich in points. In particular the pigeonhole seed ("energy KN^2 forces a carrier with $\geq K$ pairs") promotes to "forces a circle containing $\geq 2K$ pattern points", whose INTERNAL energy the universal single-circle theorem caps at $14\lambda^2 I_D^2$ (any amplitudes).*

Proof. Antipodality on D is the involution $x \mapsto 2c_D - x$; an involution without fixed points on $D \setminus \{\text{center preimages}\}$ partitions points into disjoint 2-cycles; each pattern pair carried by D is one such 2-cycle of pattern points. Disjointness and the exact count follow; the cap is the v1.8 universal theorem verbatim. $\square$

Lemma 2 (L2: extraction decomposition – exact bookkeeping). *Let $\mathcal{C} = \{C_1, \dots, C_m\}$ be any family of circles. Call an ordered pair (u, v) HOMO if some $C_s \in \mathcal{C}$ contains both u and v with $\text{carrier}(u, v) = C_s$, else HETERO. Then, by the antipodal-carrier partition (every pair has exactly one carrier),*

$$\sum_{t \neq 0} w_t^2 = E^{\text{homo}}(\mathcal{C}) + E^{\text{het}}(\mathcal{C}), \quad E^{\text{homo}}(\mathcal{C}) \leq 14\lambda^2 \sum_{s=1}^m I_{C_s}^2,$$

where I_{C_s} is the intensity of the pattern points on C_s , and E^{het} is the carrier-partition energy of the hetero pairs — the SAME object one level down, restricted to pairs not internal to any family member.

Proof. The partition is the v1.9 theorem (each pair has exactly one carrier); the homo cap is the universal single-circle theorem applied per family member to its internal pairs (a pair internal to C_s has C_s as its carrier — both points on C_s and antipodal on it). The split is exhaustive and disjoint by definition. $\square$

Lemma 3 (L3: dichotomy step – conditional). *Fix a threshold K . Either every hetero-pair carrier has $p_D^{\text{het}} < K$, in which case*

$$E^{\text{het}} \leq K \cdot \max_D w\text{-mass per pair} \cdot (\text{hetero pair } \ell^1) \leq K \lambda'^2 c_{\ell^1} I S^2\text{-form}$$

(ℓ^1 -controlled by Lemma C' of the main chain), or some carrier D^ has $p_{D^*}^{\text{het}} \geq K$, and by L1 D^* contains $\geq 2K$ pattern points: adjoin D^* to $\mathcal{C}$, strictly converting its internal hetero pairs to homo.*

3. The named obstruction (honest)

DR2-SHARE: the iteration of L3 grows the family, but family members may SHARE pattern points (a point can lie on many circles). Then $\sum_s I_{C_s}^2$ is NOT bounded by I^2 (subadditivity fails under sharing: a point contributing to k circles is counted k times), so the L2 homo cap degrades with the sharing multiplicity. This is the exact sticking point: bounding the maximum point-circle sharing within the adjoined rich-circle family is an incidence statement of precisely the class assessed research-grade (adjacent to the open circle-incidence conjecture — the verdict-#14 record's assessment, here pinned to a single named quantity: $\chi(P) := \max_x \#\{s : x \in C_s, |C_s \cap P| \geq 2K\}$). The elementary route closes iff $\chi(P) = O(\text{polylog})$ class-wide; the chain's measured worst families (exponents 2.04–2.08) are consistent with bounded χ , which is the pre-registered sharp- $O(N^2)$ conjecture seen from the sharing side.

4. Programme status

DR-2 remains RESEARCH-GRADE and OFF the critical path (verdict #14). The note's value: the route is now a three-lemma architecture with ONE named obstruction (DR2-SHARE), replacing the previous one-line seed. Any future attack has its target quantity (χ) and its exact failure mode on record.

5. Numerical verification

No new machine numbers; L1/L2 are exact structural statements over the certified carrier-partition machinery (v1.9: ℓ^1/ℓ^2 reconstructions exact to $10^{-12}/10^{-15}$). A registered spot-check: measure $\chi(P)$ on the chain's worst families (rand, ring unions, AP stacks) — expected $O(1)$, consistent with the measured exponents.

Quantitative sanity check (mandatory): consistency — L1 applied to the single-circle case ($m = 1$, all pairs homo) reproduces the universal theorem with zero hetero remainder, and applied to the two-ring family reproduces the exact $c_{\text{ring}}(n)$ accounting (the five-orbit pairs are all homo on the two member circles plus the axial transfers, which the closed form counts); limit case — at $K > n/2$ no carrier can be rich (L1 would demand $> n$ points) and L3's first branch is automatic, recovering the trivial $E \leq K$ -linear bound; sign-direction — adjoining circles only moves energy $\text{het} \rightarrow \text{homo}$ (never creates energy), so the iteration is monotone in the bookkeeping, and the obstruction concerns only the CAP's growth, stated honestly.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) “L3's first branch quotes an ℓ^1 -controlled hetero bound without a sharp constant; as stated it is not yet a usable inequality.” VALID-with-mitigation: correct — L3 is labelled CONDITIONAL

and its first branch's constant is deliberately schematic (the Lemma-C' ℓ^1 cap gives A form, but the sharp pairing with the per-pair w -mass needs the weighted version); the note's claim is the ARCHITECTURE plus the exact L1/L2, with L3's constant chase as the registered next step IF the operator ever puts DR-2 on path.

(β) *"The obstruction framing could be wrong: maybe the iteration can avoid sharing by choosing disjoint sub-families (greedy disjointification)."* VALID-with-mitigation: greedy disjointification (keep only point-disjoint rich circles) preserves the homo cap but WEAKENS the conversion (pairs internal to a discarded rich circle stay het), so the dichotomy no longer terminates — this trade is exactly why the obstruction is genuine and is recorded as such; a cleverer selection rule is the open content.

(γ) *"Pinning DR-2's difficulty to $\chi(P)$ may be an over-reduction: the open incidence conjecture has more facets than point-circle sharing."* VALID-with-mitigation: the reduction claim is scoped — "the ELEMENTARY (extraction) route closes iff χ is polylog-bounded" — not that the conjecture equals the χ bound; other routes (polynomial method, etc.) are untouched. The scoping is in the text.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / dr2-extraction-lemmas (unit U13)
Precise statement:	(L1 EXACT) a carrier with p_D pattern pairs contains exactly 2 p_D pattern points (rich carrier = rich circle). (L2 EXACT) for any circle family, $E = E_{\text{homo}} + E_{\text{het}}$ with $E_{\text{homo}} \leq 14 \lambda^2 \sum_s I_{\{C_s\}}^2$. (L3 CONDITIONAL) dichotomy: poor carriers $\Rightarrow$ K-linear het bound; rich carrier $\Rightarrow$ adjoin a $\geq 2K$ -point circle. NAMED OBSTRUCTION DR2-SHARE: point-sharing breaks $\sum I_s^2 \leq I^2$; target quantity $\chi(P)$ defined; elementary route closes iff $\chi = 0$ (polylog) class-wide.
Scope:	Unrestricted single-shell class; pure carrier-partition combinatorics; OFF critical path (verdict #14).
Dependencies:	Carrier partition theorem (v1.9); universal single-circle theorem (v1.8); Lemma C' (l1 cap); DR-2 seed record.
Evidence grade:	ANALYTIC (L1/L2 exact; L3 conditional) at T3.
Reproduction command:	(registered spot-check) measure $\chi(P)$ on the chain's worst families.
Expected output:	$\chi = 0(1)$ on rand/ring/AP families; JSON under claims/B5-BEYOND-LAYER-BOUND/runs/.
Falsification gate:	A family with χ growing polynomially AND energy exponent approaching the $\sqrt{n}$ ceiling (would mean the elementary route is

dead AND the sharp conjecture is threatened
-- honest negative pre-registered).

Tier before / after: No tier action; DR-2 stays research-grade
off-path; B5 stays T5-CANDIDATE.

No-overclaim statement: Nothing here closes or advances the
unrestricted-class gate; L3's constant is
schematic by design. PDF build + linter
PENDING operator-side.

Next required action: (only if operator puts DR-2 on path) L3
constant chase + chi spot-check. Mainline
continues (U14).